

# Toward Trustworthy FPGA Trojan Localization from Configuration Bitstreams

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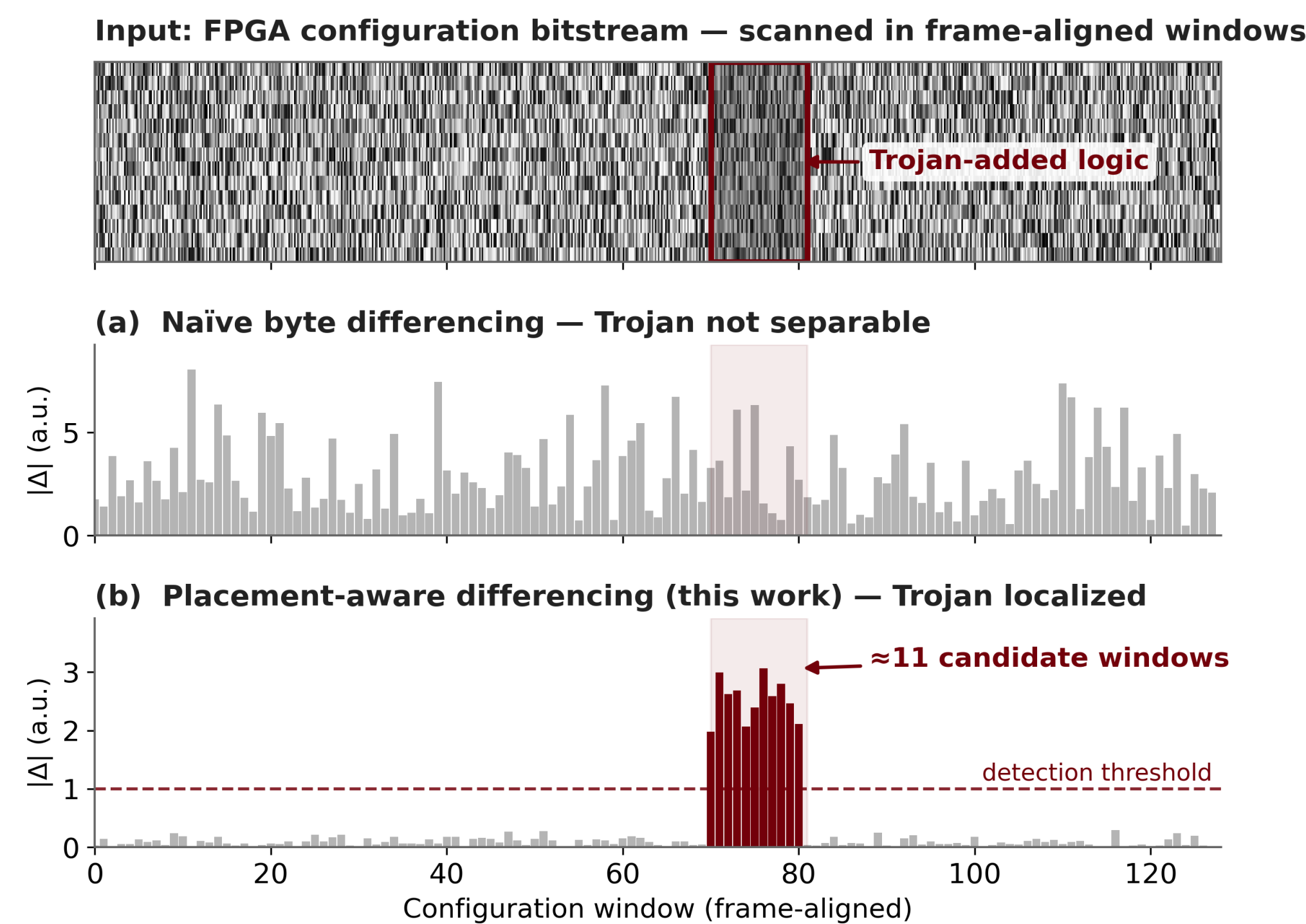
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## Motivation

- Machine learning can detect suspicious FPGA configuration bitstreams, but deployment requires more than binary detection.
- Once a bitstream is flagged as malicious, a security analyst must determine **where** the Trojan resides to support verification, forensic analysis, or deployment decisions.
- A straightforward approach is to compare trusted and suspicious bitstreams and attribute the largest differences to the Trojan.
- However, implementation artifacts introduced during synthesis, placement, and routing can dominate these differences, making naïve localization unreliable.
- This work investigates the implementation conditions required for **trustworthy FPGA Trojan localization**.

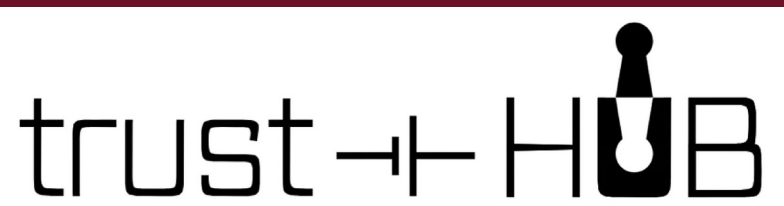
## How Localization Works



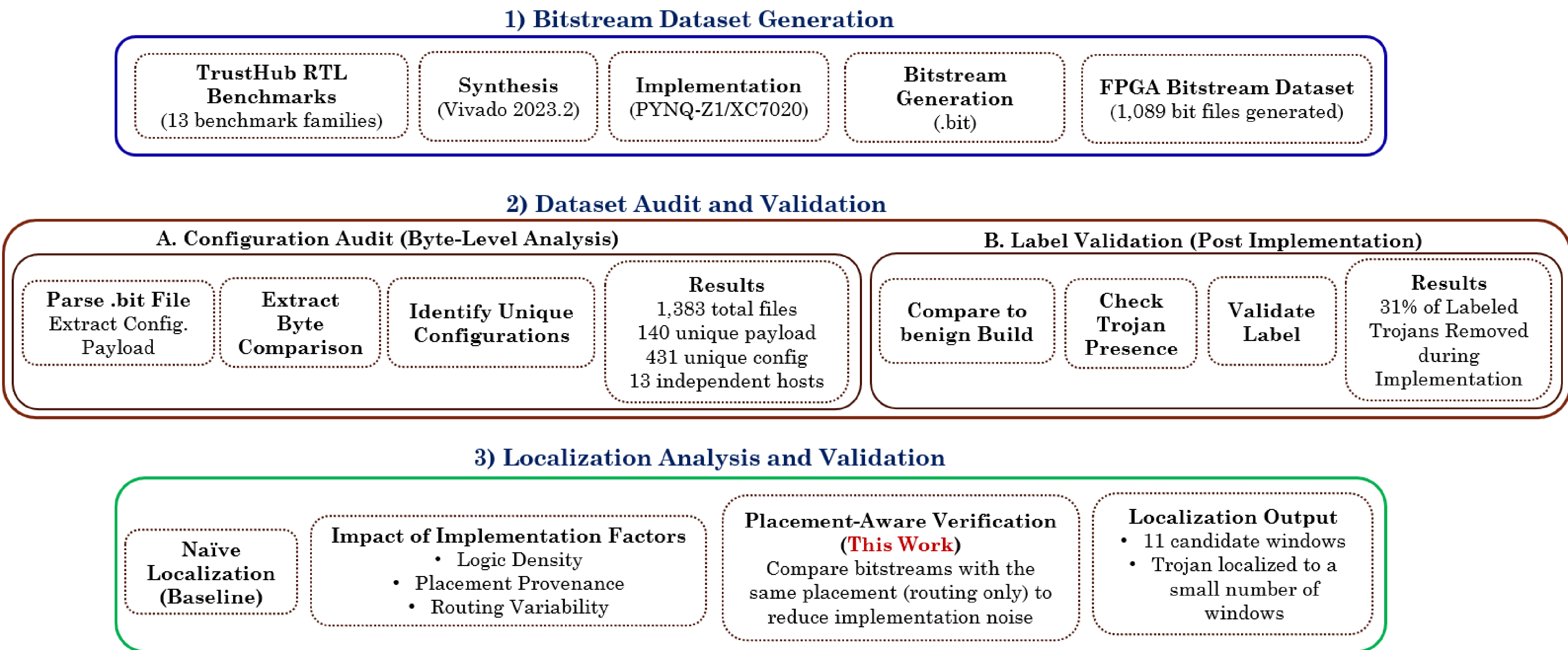
We scan the input bitstream in frame-aligned windows. Naïve byte differencing is high across the whole file, so the Trojan is not separable; placement-aware differencing (comparing two builds that share placement) cancels routing/layout noise, so only the Trojan-added windows survive — localizing it to ~11 candidate windows. (Illustrative signal; window count matches the measured result.)

## References

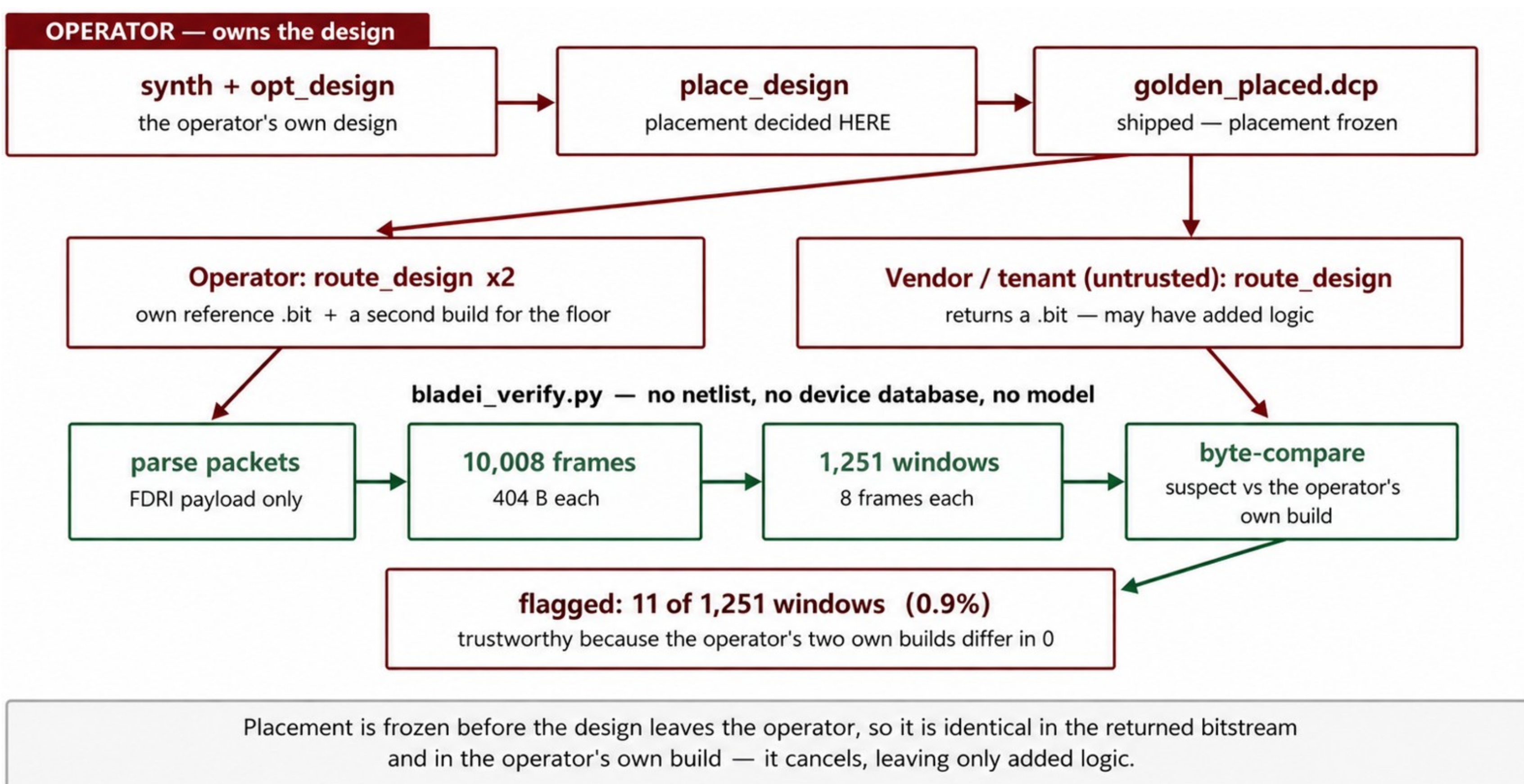
- Stahle-Smith, R., Antley, C., Bakos, J. D., & Karakchi, R. (2026). Hardware-accelerated line-rate bitstream screening for secure FPGA reconfiguration. arXiv preprint [arXiv:2605.08984](https://arxiv.org/abs/2605.08984).
- R. Stahle-Smith and R. Karakchi, "From Bits to Trust: AI-Driven Reconfigurable Security," SoutheastCon 2026, Huntsville, AL, USA, 2026, pp. 1-6, doi: 10.1109/SoutheastCon63549.2026.11476474.



## Methodology



## Placement-Aware Verification



## Experimental Results

| Experiment                   | Result                 | Conclusion                                     |
|------------------------------|------------------------|------------------------------------------------|
| Dataset audit                | 1,383 → 431 → 140 → 13 | Effective diversity is substantially reduced.  |
| Label validation             | 31% invalid labels     | Labels require post-implementation validation. |
| Naïve localization           | Density ≠ Trojan       | Bitstream differences alone are unreliable.    |
| Placement-aware verification | ~11 candidate windows  | Practical deployment-time localization.        |

## Objective

- How can FPGA hardware Trojans be localized reliably from configuration bitstreams despite implementation-induced variability?

## Key Findings

| Observation                 | Finding                                                  |
|-----------------------------|----------------------------------------------------------|
| Dataset collapse            | Implementation reduces effective dataset diversity.      |
| Invalid labels              | RTL labels may not survive synthesis and implementation. |
| Naïve localization          | Detects logic density rather than Trojan location.       |
| Place-and-route variability | Implementation noise masks Trojan-induced changes.       |
| Shared placement            | Enables trustworthy FPGA Trojan localization.            |

## Dataset and Audit Results

| Category             | Metric                                         | Result                                  |
|----------------------|------------------------------------------------|-----------------------------------------|
| Dataset              | Design families                                | 13                                      |
|                      | Generated bitstreams                           | 1,089                                   |
|                      | Benign / Malicious                             | 545 / 544                               |
|                      | Unique configuration payloads                  | 140                                     |
| Configuration Audit  | Duplicate payloads                             | 949 duplicate implementations*          |
| Label Validation     | Malicious bitstreams with no observable Trojan | 169 (31%)                               |
| Evaluation Integrity | Configuration overlap                          | Present across common train-test splits |
|                      | Metadata leakage                               | Identified                              |

## Acknowledgment

- This work was supported by the McNair Junior Fellowship and the Office of Undergraduate Research at the University of South Carolina.
- Benchmark designs were sourced from Trust-Hub (NSF).
- Hardware and tools were provided via the AMD University Program.
- ChatGPT assisted in drafting, editing, formatting prose, and generating figures; all technical content, experiments, and analysis were performed solely by the author, who assumes full responsibility for the complete work.
- We acknowledge the foundational contributions of Rye Stahle-Smith during his undergraduate studies